

Assessment of a 15-year white-tailed deer management program and woody recovery in a suburban forest preserve

Christopher Nagy^{a,*}, Chloe Ng^a, Norman Veverka^a, Mark Weckel^b

^a Mianus River Gorge, Inc., 167 Mianus River Rd, Bedford, NY 10506, USA

^b Center for Biodiversity and Conservation, American Museum of Natural History, 200 Central Park West, New York, NY 10024-5102, USA

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ABSTRACT

Overbrowsing by highly abundant white-tailed deer (*Odocoileus virginianus*) negatively impacts the regeneration of the forest understory throughout the eastern United States. Deer management programs are often used by land managers to reduce deer densities and promote forest regeneration. This study examines the impact of a 15-year-long archery-based deer management program on the understory of a nature preserve in southern NY consisting of an old-growth hemlock and second-growth mixed hardwood forest. Vegetation plots were sampled every 3–5 years between 2004 and 2019, while deer density was estimated using camera trap survey data annually since 2009. Generalized mixed-effects models were used to determine the influence of deer density, forest type, canopy cover, and time on species richness, native stem density and total stem density. The most predictive models showed an increase in richness as time went on. Additionally, stem density had a negative relationship with deer density (i.e., stem density and species richness increased as deer density declined over time). For both native and total stem density, the highest-ranked model included only deer density as a predictor. Generally, woody species richness and stem density increased in the seedling (<0.3 m tall) and sapling (0.3–0.9 m tall) size classes but remained relatively constant for the transgressive (>0.9 m) size class over time and as deer density decreased. Many native species exhibited density increases in the smallest size classes, but black birch (*Betula lenta*), American beech (*Fagus grandifolia*), and mapleleaf viburnum (*Viburnum acerifolium*) were the only native species to increase in the transgressive (>0.9 m tall) class. Several non-native species, however, did increase in the transgressive size class since 2004. These results indicate that the forest has improved as a result of the DMP, but long-term recruitment of native species is still uncertain. The DMP has shown the potential to help restore woody regeneration but monitoring should continue to ultimately determine whether deer management can allow for true recruitment of native seedlings into the forest understory and canopy.

1. Introduction

The impact of abundant white-tailed deer (*Odocoileus virginianus*; hereafter, deer) on forests in the northeast US (Côté et al. 2004; Blossey et al. 2019; Royo and Stout 2019) and elsewhere (Tremblay et al. 2007, Tanentzap et al. 2011, Redick and Jacobs 2020) is well documented. Deer cause declines in survival, growth, and overall regeneration of

many woody (Jenkins et al. 2015, Bradshaw and Waller 2016, Blossey et al. 2019, Ward and Williams 2020) and herbaceous (Balgooyen and Waller 1995, Rooney and Waller 2001, Pendergast et al. 2016) plants while simultaneously facilitating the prevalence and spread of invasive species (Baiser et al. 2008, Duguay and Farfaras 2011, Averill et al. 2016). Ecological stability, diversity, growth, and productivity of forests can be reduced by superabundant deer via a host of direct and indirect

Abbreviations: DMP, deer management program; MRG, Mianus River Gorge, Inc.; NYSDEC, New York State Department of Environmental Conservation; DMAP, deer management assistance program; DDP, deer damage permit; SECR, spatially explicit capture recapture; Spikes+4pts, demographic group that pools younger bucks, usually around 1.5 years old, with non-branching antlers and/or fewer than 5 total branch tips (“points”); AIC_c, Akaike’s Information Criterion corrected for small sample size; w_i, Akaike weight for model *i*; C, vegetation plots categorized as “coniferous”; i.e., within the coniferous-dominated forest; H, vegetation plots categorized as “hardwood”; i.e., within the deciduous-dominated forest; DD, predictor variable used in modeling representing deer density in deer/km² for a given year; Time, predictor variable used in modeling representing years zeroed at 2004; i.e., number of years since deer management began; CC, predictor variable used in modeling representing canopy closure in percentage closed scaled to a mean of 0 and standard deviation of 1.

* Corresponding author.

E-mail address: chris@mianus.org (C. Nagy).

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mechanisms (Côté et al., 2004, Bressette et al., 2012, Shelton et al. 2014, Lesser et al. 2019). In suburban areas, these effects can become especially intense because of a lack of natural predators, lower hunting prevalence, limited hunter access to land, and anthropogenic food subsidies – both direct (e.g., winter feeding) and indirect (e.g., gardens, lawns) – that can subsidize deer through times when wild food is scarce (DeNicola, 2000, Kilpatrick and Spohr 2000, Etter et al. 2002, Urbanek and Nielsen 2013).

Many municipalities, land trusts, parks, and other open space organizations have instituted Deer Management Programs (DMPs) that attempt to reduce their local deer populations (e.g., Kilpatrick et al. 2014, Blossey et al. 2019, deCalesta and Eckley 2019). In addition to enhancing forest health, DMPs can be enacted to reduce deer-vehicle collisions, reduce the prevalence of tick-borne diseases, and limit property damage caused by deer (e.g., damage to residential gardens, agriculture, etc.; Conover and Kania 1988, deCalesta and Eckley 2019). Our assessment in this study focuses on managing deer to reduce herbivory pressure and thereby restore native biodiversity and woody regeneration.

The most common method of deer management employed by local or private DMPs is to allow recreational hunters, under varying levels of administrative oversight, access to land (see deCalesta and Eckley (2019) for several case studies). Many private land trusts and nature preserves now instituting DMPs historically prohibited hunting of any kind. Other models include “culls” where a DMP employs private contractors or government officials to harvest large numbers of deer using means not permitted to the recreational hunter for safety and/or ethical reasons, such as the use of bait, hunting at night, using suppressed rifles, etc. Regardless of the harvest method, DMPs are often limited to certain areas or parcels for a variety of reasons. In addition, suburban DMPs that employ recreational hunters may work under additional restrictions on hunting or firearm use based on county or local laws. Limitations on land access, weapons, season, and spatial scale may make deer management simply ineffective, in terms of not reaching the necessary ecological impact and/or from the standpoint of cost in time and resources required to manage the DMP compared to other stewardship needs.

Fundamentally, the goal of any biodiversity-focused DMP is to enact a three-step process. First, the DMP managers seek to employ some method, usually recreational hunting, to efficiently harvest a substantial number of deer and thereby lower the density of deer on their property. This reduction of deer numbers, in turn, ought to lead to reduced impacts of herbivory by deer upon the vegetation, in turn allowing regeneration of the impacted trees, shrubs, and wildflowers. While this cause-and-effect process is ecologically plausible, any one of the steps may not occur in practice and/or the cost in time and resources may turn out to be too high to sustain the DMP over the long-term. The most important part of the process to monitor is the last and ultimate objective: do the plants grow back? Even if hunting substantially reduces local deer density, this may not lead to vegetative recovery: deer may increase their individual rates of herbivory at lower densities, or other ecological or historical factors (e.g., degraded seed bank, legacy effects, canopy closure, erosion, invasive species) may emerge as new limiting factors (Mladenoff and Stearns 1993, Tanentzap et al. 2011).

Hence, if resources to do so are available, monitoring the other components of the process in addition to the primary goal of vegetative recovery is still valuable; assessing hunting efficacy and monitoring deer dynamics can help determine where the process is breaking down if vegetative recovery does not occur. Hunting may or may not lead to a meaningful harvest, or it may be too much work for hunters and/or managers to implement. Or, harvest may be substantial but deer density may not decrease: immigration from adjacent lands or increased reproduction might offset any attainable level of harvest. Hunting efficacy is likely the easiest to measure and provides important data for administering the program (e.g., which hunters are most productive, what is the average hours per harvest rate, etc.). Absolute or relative deer density can be a useful short term measure of progress compared to

vegetative regeneration which could have intrinsic time lags and is determined by many complex, interacting processes including herbivory (Tanentzap et al. 2011, Lesser et al. 2019).

The Mianus River Gorge, Inc. (MRG), a land trust located in Westchester County, NY, implemented a DMP on their main parcels totaling approximately 323 ha in 2004 (see Weckel and Rockwell (2013) for more detail) and since then has kept records of hunter effort and harvest. The staff also established 22 large vegetation plots that would be maintained and sampled to monitor long-term changes in the woody and herbaceous plant community and, since 2009, has conducted annual deer density surveys. The purpose of this study was to assess the MRG's DMP at the three levels described above, as a case study representing the potential for archery-based deer management in suburban/exurban areas to improve native tree and shrub regeneration. We paired local and regional estimates of deer density with MRG's own deer estimates to see the extent of deer reduction due to the harvest effort. Second, we examined changes in the vegetative community and modeled the relationship between changes in deer density and changes in species richness and stem density of the woody plant community on the MRG Preserve over a 15-year period, 2004 to 2019. Overall patterns illuminated by modeling were then examined more closely with regard to stem size and species. Lastly, we compared our most recent sample in 2019 to the comparable plots and study area of prior surveys of the Preserve in 2004 (Weckel et al. 2006) and 1965 (Bard 1967) surveys to examine long-term change in the understory of the old growth hemlock stand in the interior of the MRG Preserve.

2. Material and methods

2.1. Site description

The MRG Preserve is, as of 2019, a 374 ha nature preserve located along the Mianus River in Westchester County, NY and Fairfield County, CT. Approximately 186 ha along the river is old-growth eastern hemlock (*Tsuga canadensis*), while the remainder primarily consists of post-agricultural second-growth mixed hardwood forest (Fig. 1). The immediately adjacent landscape is exurban development (a typical residential lot is 1.6 ha), while northeastern Westchester and western Fairfield Counties overall could be characterized as suburban. The DMP focuses on the core parcels of the MRG Preserve, all of which are in New York State; little-to-no deer management is performed on MRG's lands in Connecticut.

2.2. Summary of the deer management program

MRG's management program began in 2004 and has largely relied on recreational hunters following standard regulations enacted by the New York State Department of Environmental Conservation (NYDEC), with a few exceptions described below. The normal deer season in New York State runs from October 1 through December 31. In other parts of the state, there are separate seasons of varying lengths that allow for hunting with vertical (compound) bow, crossbow, and firearms, but in Westchester County deer hunting is permitted only with a vertical bow. Baiting deer is prohibited in New York State, and hunting is allowed during the daylight hours only. Hunting is permitted seven days a week.

Beyond these ‘normal’ regulations, New York State allows landowners dealing with negative deer impacts, including biodiversity loss in natural areas due to overbrowsing, to apply for Deer Management Assistance Program (DMAP) tags. DMAP tags are antlerless-only tags licensed to the landowner, who is then empowered to use the tags themselves or to distribute the tags to hunters using the permittee's land during the normal October through December deer season to make additional harvest, up to four per hunter. Additionally, landowners and managers can apply for Deer Damage Permits (DDP) that can allow for harvest outside of the normal season, the use of bait, harvest at night, and/or the use of shotguns or crossbows. MRG used DMAP permit tags

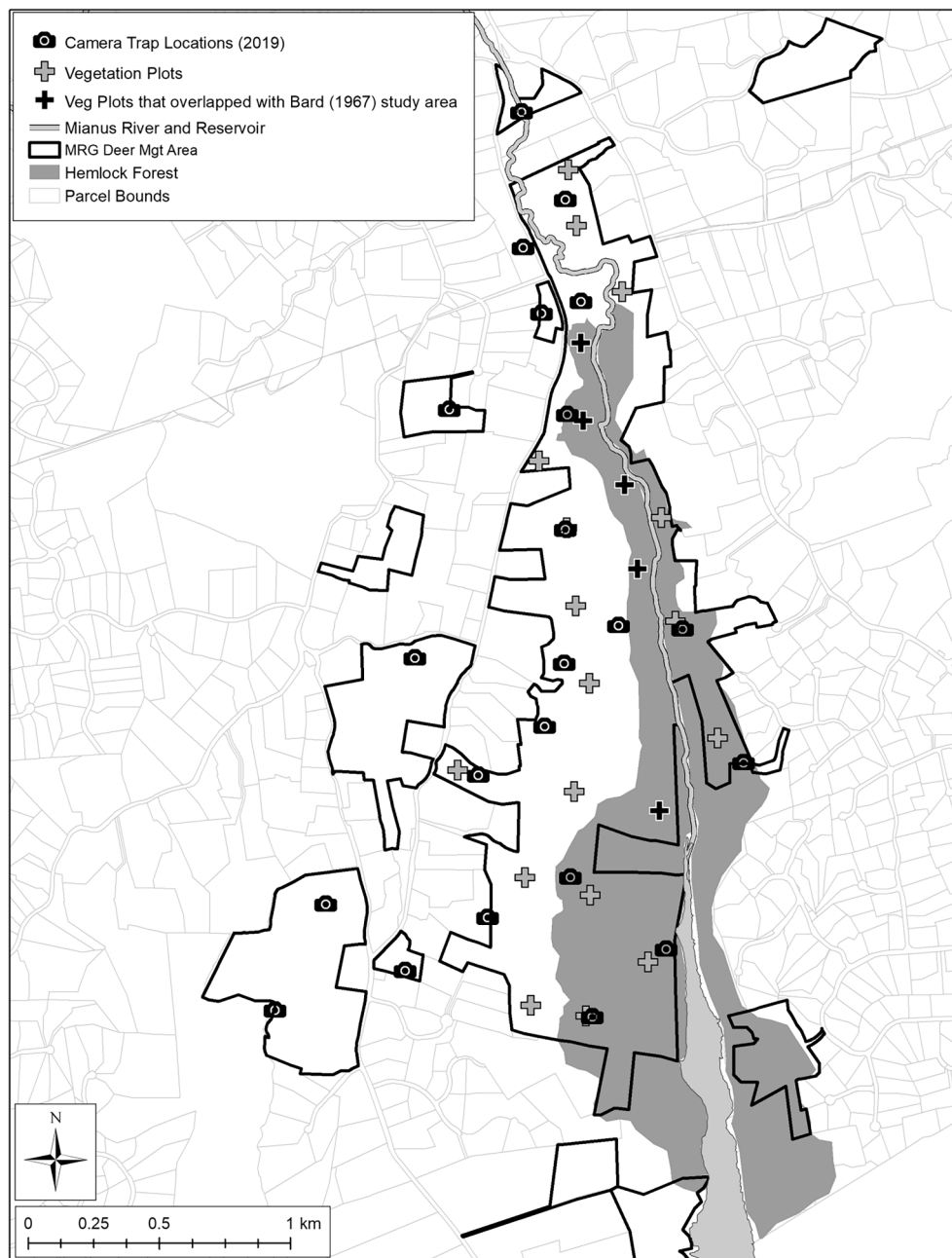


Fig. 1. Locations of vegetation plots (crosses) and typical camera trap locations used in annual deer surveys (camera icons) in the MRG Preserve. Shaded areas represent eastern hemlock forest; other sections are second-growth hardwood forests of maple-beech-birch or oak-hickory. Parcel boundaries of the surrounding neighborhood (grey lines) illustrate the exurban landscape.

every year, and used DDPs from 2004 to 2008, 2010, and 2015 to the present (Table 1). In the earlier instances, MRG used bait in a limited and haphazard way, did not hunt at night, and continued to rely on their volunteer hunters to increase harvest under their DDPs. Since 2016, the DDP period has been more coordinated, where staff established bait stations, monitored them through the harvest period, and hunted them systematically (along with select volunteer hunters) at dawn, dusk, and night. Crossbows were also permitted and used during this DDP period as they tend to be more accurate and can reliably be shot from farther away than vertical bows. MRG has been hesitant thus far to use shotguns in the Preserve due to community relations concerns.

Between five and 23 hunters (Table 1), varying among years, were permitted to hunt on the main parcel and a few nearby parcels owned and/or managed by MRG (Fig. 1). The Preserve was not open generally

to any licensed hunters; the DMP hunters were interviewed and briefed on the conservation mission of the program and each had to pass an archery test before they were accepted into the program. Hunters had to take the archery test every 3 years. The cohort of participating hunters remained relatively stable through the 15 years with a few removed or a few new people accepted as needed. Hunters were encouraged to harvest does over males and fawns, as removing adult females has a greater impact on reproductive potential and population growth than removing males in a polygynous species such as white-tailed deer (Turner et al. 2016). In addition, hunters had to “earn-a-buck,” meaning they had to harvest two does for each mature (trophy) male they harvested. This incentivized doe harvest while still providing a way for hunters to obtain a trophy buck, which is one of the primary motivations for most hunters to participate (Weckel et al. 2011b). Total area changed from year to

Table 1

Annual hunting seasons, number of hunters, management area (land open to hunting), total hours, total harvest (doe harvest: females > 1 year old), hour/harvest, and harvest/ha of the DMP, Mianus River Gorge Preserve, Bedford, NY, 2004 – 2019. Asterisks indicate post season harvest periods 2016 and after that were coordinated and implemented by MRG staff using crossbows, bait stations, lights, and nighttime hunting.

Year	Harvest Season	Mgt. Area (ha)	Hunters	Total Hours	Total Harvest (Doe Harvest)	Injured/Not Recovered	Hours/Harvest	Harvest/ha
2004–2005	Reg.: Oct 1 - Dec 31	92	5	279.75	16 (16)	1	17.48	0.38
	Post: Jan 1 - Feb 29		4	90.5	19 (17)	0	4.76	
2005–2006	Reg.: Oct 1 - Dec 31	153.3	9	600.33	20 (16)	7	30.02	0.18
	Post: Jan 1 - Feb 28		4	87.5	7 (7)	4	12.5	
2006–2007	Reg.: Oct 1 - Dec 31	166	10	407	26 (17)	0	15.65	0.17
	Post: Jan 1 - Feb 28		4	29.65	3 (3)	1	9.88	
2007–2008	Reg.: Oct 1 - Dec 31	172.3	23	742.75	27 (19)	1	27.51	0.16
	Post: Jan 1 - Feb 28		13	131.67	1 (1)	0	131.67	
2008–2009	Reg.: Oct 1 - Dec 31	183.2	16	837.75	22 (16)	2	38.08	0.12
	Post: Jan 1 - Feb 28		5	70	0 (0)	0	X	
2009–2010	Reg.: Oct 1 - Dec 31	264.5	18	1114.6	29 (26)	5	38.43	0.11
2010–2011	Pre: Sept 1 – Oct 1	191.4	9	174.92	7 (6)	2	24.99	0.18
	Reg.: Oct 1 - Dec 31		16	652.75	27 (20)	2	24.18	
2011–2012	Reg.: Oct 1 - Dec 31	174.5	13	595.25	11 (9)	1	54.11	0.06
2012–2013	Reg.: Oct 1 - Dec 31	286.8	11	454.5	6 (5)	2	75.75	0.02
2013–2014	Reg.: Oct 1 - Dec 31	248.8	11	609.25	23 (18)	3	26.49	0.09
2014–2015	Reg.: Oct 1 - Dec 31	290.1	12	701.13	14 (11)	0	50.08	0.05
2015–2016*	Reg.: Oct 1 - Dec 31	214.9	11	557.08	7 (6)	4	79.58	0.05
	Post: Jan 1 - Mar 31		7	171.17	3(1)	0	57.06	
2016–2017*	Reg.: Oct 1 - Dec 31	240.8	11	457.25	8 (7)	2	57.16	0.05
	Post: Jan 1 - Mar 31*		2	99.75	3 (1)	1	33.25	
2017–2018*	Reg.: Oct 1 - Dec 31	280.8	13	377.42	13 (9)	1	29.03	0.07
	Post: Jan 1 - Mar 31*		2	89.43	7 (4)	1	12.78	
2018–2019*	Reg.: Oct 1 - Dec 31	263.3	11	410	7 (5)	1	58.57	0.05
	Post: Jan 1 - Mar 31*		3	57.58	7 (3)	0	8.23	
2019–2020*	Reg.: Oct 1 - Dec 31	282.9	10	692.65	21 (17)	0	32.98	0.10
	Post: Jan 1 - Mar 31*		5	183.83	7 (5)	1	26.26	

year as MRG acquired and opened new parcels for deer management. Hunters were required to submit logs of every hunt they performed on MRG land. These included location and time of the hunt, deer observed sorted by demographic group (mature males with branching antlers (i. e., bucks), males of approximately 1.5 years of age with small, non-branching antlers (i.e., spikes), females > 1 year old (does), and young deer < 1 year old of either sex (fawns)), any arrows shot, and any deer harvested. MRG staff validated any harvest via meeting the hunter in the field (preferred) or photograph submitted by the hunter of the carcass and tag. Hunting logs and harvest records were used to calculate summary statistics such as total harvest, antlerless harvest, total hours, hours per antlerless harvest, and hunter-specific statistics.

MRG set a target deer density of 5.8 – 6.9 deer/km², based on [Tilghman's \(1989\)](#) research that examined the effects of deer at varying densities upon the regeneration of common hardwood species in northwestern Pennsylvania. MRG prioritizes the preservation of its eastern hemlock forest, and existing research on the impact of super-abundant deer on hemlock specifically recommend even lower densities – between 2 and 6 deer/km² ([Alverson et al. 2019](#)) – but at the start of its DMP, MRG aimed for the more general mixed hardwood recommendations from Pennsylvania by [Tilghman \(1989\)](#). Even this less ambitious target would require a reduction of 66% as the non-hunted deer density was approximated as 28 deer/km² based on surveys performed elsewhere in the region ([Table 2](#)).

2.3. Deer density estimation

2.3.1. Estimating deer density with camera traps

Generally, precise estimates of abundance or density are difficult to obtain for wildlife species that lack unique marks, either natural or artificial. Trapping and marking deer is typically beyond the resources or expertise of land trusts or municipal agencies and doing so in a suburban environment is often infeasible due to property access. A few methods have been developed that use the branching patterns of the antlers of mature male deer (bucks) as individually distinctive markings. [Jacobson et al. \(1997\)](#) used camera traps to survey deer and used these

Table 2

Deer surveys of unmanaged herds in southern NY and CT used to establish 2004 unmanaged deer density at the Mianus River Gorge Preserve prior to DMP implementation, Bedford, NY, 2004.

Location	Year	Dist. from MRG (km)	Deer/ km ²	Method	Source
Bedford, NY	1999–2001	2.8	22	Spotlight surveys	Daniels et al. 2009
Ward Pound Ridge Preserve, Pound Ridge, NY	2008	8.46	24.6	Pellet counts	deCalesta and Pierson 2008
Lewisboro, NY	1999–2001	10.2	27.7	Spotlight surveys	Daniels et al. 2009
Teatown Lake Reservation, Ossining, NY	2009–2013	17.43	28.1	Pellet counts	D. Begley- Miller, unpub. data
Greenwich, CT (rural)	2004	17.55	26.3	Aerial and spotlight surveys, hunting data,	Brash et al. 2004
Greenwich, CT (suburban)	2004	17.55	20.2	telemetry Aerial and spotlight surveys, hunting data,	Brash et al. 2004
Groton, CT	1995–1999	130.57	46.9	telemetry Aerial surveys	Kilpatrick et al. 2014

branching antler patterns as recorded in the photographs to count bucks. The ratio of individual bucks to the total number of buck observations was used to extrapolate the abundances of the non-identifiable

demographic groups – adult females (does), fawns of both sexes, and immature males (spikes) – from the respective total observations of each. Jacobson's method assumed that the observed ratio of buck abundance to buck observations is universally applicable to other demographic groups; i.e., this ratio is the same for doe abundance to doe observations, and fawns to fawn observations, etc. This assumption allows for extrapolation from the buck data to the other demographic groups.

Weckel et al. (2011a) built upon Jacobson's method to account for potential variation in 'detectability' or capture rate among these demographic groups by regressing, for each demographic group, total observations versus the daily rate of recording at least one photographic observation (i.e., trap success) for each camera. These linear slopes were then used as "x-to-Buck" ratios that provided more appropriate conversions from the known buck abundance to the unknown doe/fawn/spike abundances. Weckel et al. (2011a) also provided a method to estimate precision of these abundance estimates by bootstrapping subsamples of cameras, repeating the regression with each subsample, and building prediction intervals from the resulting bootstrap of demographic ratios. However, a few areas of improvement remained. Both Jacobson (1997) and Weckel et al. (2011a) used the raw count of bucks from the photographs in their protocols, rather than incorporating some correction for imperfect capture rate, e.g., via a mark-recapture analysis. Also, the Jacobson and Weckel et al. estimators both calculate a final herd abundance which then needs to be converted to a density via an ad hoc judgement of the study area.

Since its publication, Weckel et al.'s (2011a) estimator has been further developed by MRG staff to incorporate a spatially explicit capture recapture (SECR) analysis (Efford et al. 2009a, Efford et al., 2009b) using buck detection histories. This provides more accurate estimates for the buck census and explicitly estimates density rather than abundance. SECR also provides confidence limits for buck density which are incorporated into the prediction intervals of the other demographic groups as well. In addition, while Weckel et al. (2011a) used the total number of observations across the entire survey (termed "RPO" in that study), we used mean observations per day. This change allowed the direct comparison of cameras that may have run for different lengths of time due to equipment malfunction or slightly different start and end dates (it generally took 1 person > 1 day to deploy or collect all cameras).

Unfortunately, neither deer abundance nor density were estimated prior to 2009 in the MRG Preserve itself. An initial density estimate for 2004 – the start of both the DMP and vegetation monitoring (see below) – was necessary to properly examine the change in deer and any subsequent change in vegetation. Using six other surveys conducted in the region (Table 2), we estimated a likely starting density of 28.0 ± 8.84 deer/km² (mean $\pm$ SD) in the unhunted deer herd in the MRG Preserve.

2.3.2. Camera survey protocol

Since 2009, MRG staff conducted camera surveys to estimate deer density on the Preserve using Reconyx HC500 and PC800 camera traps (Table 3). Dates of these surveys varied slightly across years but all were

performed between August and October annually. Surveying during this period allows for full growth of the bucks' antlers by the late summer but is still prior to the hunting season and the rut, where deer movements and local abundance may change frequently. In 2009, cameras were randomly located with a minimum separation of at least 300 m and no more than 500 m between cameras, based on Porter et al.'s (2004) estimate of suburban doe home range of 21.4 ha. Specific locations were chosen within 25 m of the "planned" point based on ideal tree size, level ground, vegetation density, etc. Camera locations were largely maintained over the 2009 – 2019 period but shifted slightly for logistical reasons or when new land was acquired and opened to hunting (Table 1). Cameras were placed at a height of 1 m and left unbaited. We defined an observation as a sequence of pictures of the same apparent animal where two consecutive pictures remained separated by < 10 min.

Camera deployment information and photographs were imported into MS Access (Microsoft 2016) databases using the Colorado Parks and Wildlife PhotoWarehouse (Ivan and Newkirk 2016) database tools. Species and number of animals seen in every picture were recorded and deer pictures were sorted into bucks (adult males with > 4 branching antler points), does (adult females), fawns (juveniles approximately 0.5 years old of either sex), and "spikes + 4pts.". The latter class originally included just "spikes" – those male deer approximately 1.5 year old that grow only small, unbranched antlers. However, in several years we did not obtain spike observations at enough cameras to properly run the density estimator. Simultaneously, there were large numbers of small "4-pointer" buck observations every year that were also hard to individually identify. Thus, for all surveys we pooled spikes and 4-point bucks into a spike + 4pt category that represents males that tend to be both younger (~1.5-2.5 years old) and difficult to identify.

2.3.3. SECR analysis of bucks

Daily capture histories of individual bucks were built by sorting buck observations (each typically made up of many pictures) into individual bucks based on antler characteristics. We first scored each observation by the presence or absence of right and left brow tines, and then the number of branching antler points on the left and right antlers. Once these quantitative characteristics were recorded, we could use more qualitative characteristics within the smaller pools of observations from bucks that had identical antler points. These characteristics included notable bends or angles in certain antler tines, relative size of the tines, asymmetries between left and right antlers, shape of the snout, etc., and were recorded for each identified animal. SECR has no intrinsic correction or estimation of misidentification errors, i.e., errors in the assignment of identity to recaptures, and such errors can greatly alter the final estimates of interest (Borchers 2012, Borchers and Fewster 2016), so we erred on the side of discarding blurry or ambiguous observations rather than mis-identify them.

SECR is a hierarchical model with two processes: a spatial model of the animal population and a model of the detection process. Animals are assumed to live, move, forage, etc. around "activity centers" and the

Table 3

Annual survey periods, number of cameras, and total trap nights for deer surveys, Mianus River Gorge Preserve, Bedford, NY, 2009 – 2019.

Year	Survey Dates	No. Cameras	Survey Length (days)	Total Trap Nights	Total Buck Observations	Usable Observations	Prop. Used
2009	Aug 16 - Oct 16	22	61	657	87	76	0.87
2010	Aug 1 - Sept 3	19	33	290	79	72	0.91
2011	Aug 2 - Sept 3	21	32	622	103	50	0.49
2012	Aug 1 - Aug 31	20	30	629	76	71	0.93
2013	Sept 12 - Oct 15	21	33	659	142	128	0.90
2014	Aug 26 - Sept 30	20	35	720	37	28	0.76
2015	Sept 1 - Oct 8	17	37	398	32	27	0.84
2016	Sept 15 - Oct 13	15	28	393	58	39	0.67
2017	Sept 1 - Oct 2	19	31	483	110	68	0.62
2018	Sept 6 - Sept 30	21	24	476	209	153	0.73
2019	Sept 1 - Oct 4	22	33	578	50	44	0.88

study area is modeled as a Poisson spatial point process representing the distribution of these activity centers (Borchers and Efford 2008). The parameter D denotes the density of activity centers, and hence animals, on the landscape. Detectors (which can be traps, cameras, transects, etc.) are located in the study area and their respective detection rates can be modeled according to a particular detection function. The detection model typically consists of two parameters (more complex detection functions may have more) that describe the declining probability of detecting the animal as a detector's location gets farther away from an activity center. The parameter g denotes the detection probability at distance 0, i.e., the y-intercept of the detection function, and the parameter σ describes the steepness of the detection function, i.e., the shape and steepness of the curve that describes how detection probability decays as a detector is placed farther from an activity center. In addition to these basic parameters, additional parameters can be added to denote individual or groups of animals with different values of, e.g., g or σ , trends in detection over time or across animals, as well as including habitat, environmental, or sampling covariates on any of the parameters.

SECR has four main assumptions: 1) the population is closed geographically and demographically (i.e., animals do not enter or leave the area and none are born or die during the study period); 2) individual activity centers are randomly distributed and do not change during the period; 3) detections across and within individual animals must be independent of each other; and 4) the probability of detection at detectors is a function of the distance to the detector, maximizing at distance = 0 (Green et al. 2020). We were confident that we met the first two assumptions primarily based on the time of year we sampled, and the relatively brief period (~32 days with the exception of 2009) we ran the survey. White-tailed deer home ranges are fairly stable in the summer but, in males, begin to shift in mid-autumn (Tierson et al. 1985, Grovenburg et al. 2009), and mortality is relatively low in the warm seasons when food is plentiful, there is no hunting, and weather is mild. Our buck analysis met the third assumption as capture histories were based on day, so multiple observations of the same deer on the same day were condensed and subsequent detections were separated by at least a few hours, but typically a day or more. The fourth assumption was not tested explicitly but was reasonably assured as it was far easier to identify species and individual bucks from photographs taken when the animal was closer to the camera compared to farther away.

We ran four basic SECR models for each year's set of daily capture histories. We fit a "dot" model with only the basic three parameters $D(\cdot)$, $g(\cdot)$, $\sigma(\cdot)$. We fit a "learned behavior" model where an additional parameter (b) specifying a step change in base detection probability (g) after the first detection of each individual (unlikely with cameras but possible if, e.g., deer become accustomed to them after initial shyness, or approached cameras initially out of curiosity but then later lost interest). We also fit a time trend (T), where detection probability (g) changes over time across individuals. Fourth, we fit a dot model that allowed for a 2-group mixture, which compensates for unknown heterogeneity in detection by sorting individuals into two groups based on low or high detection rate (i.e., adding g_1 and g_2 estimates). These four models were run with both a half-normal detection function and an exponential detection function, totaling 8 models for each year's data.

The models were ranked with Akaike's Information Criterion corrected for small sample sizes (AIC_c; Burnham and Anderson 2002). A supported model set was chosen by examining Akaike weights (w_i) starting with the top-ranked model and including the next best model until the sum of w_i 's reached a clear dropoff point or a sum of 0.9. Buffer sizes of supported models were examined from 1 to 4 times σ to ensure that model estimates did not vary with σ . The density estimates of this set of supported models were averaged according to their respective recalculated w_i 's and used for the subsequent demographic extrapolation. SECR analysis was performed using Program R (R Core Team 2019) and the secr (Efford 2021), MASS (Venables and Ripley 2002), ggplot2 (Wickham 2016), boot (Davison and Hinkley 1997, Canty and Ripley

2021), and svDialogs (Grosjean 2021) packages.

2.3.4. Demographic extrapolation

After measuring buck density with SECR, we proceeded with the extrapolation to other demographic groups as described by Weckel et al. (2011a) and above (Section 2.3.1). This most recent version of the estimator, using observations/day instead of total observations and incorporating a SECR analysis, was used on MRG's buck photographs from 2009 to 2019. While SECR provides density estimates rather than abundance, the derivations of non-buck densities from buck density can be calculated identically to Weckel et al.'s original extrapolations of non-buck abundances from buck abundance. Total deer density was obtained by summing the SECR-estimated buck densities and the extrapolated doe, fawn, and spike + 4pt densities for each year.

2.4. Vegetation sampling and analysis

A set of 22 vegetation plots were established and sampled in 2004 prior to the beginning of MRG's DMP in October 2004. Plots were established throughout the largest and central parcel (178 ha) of the MRG Preserve, randomly located while maintaining a minimum separation of 200 m. These plots were categorized as "C" (coniferous) or "H" (hardwood) depending on which forest type (Type) they were in. All but one of the "C" type plots were in eastern hemlock stands, with the exception (C1) located in a white pine (*Pinus strobus*) stand. Plots consisted of two 60 × 1 m strips set perpendicular to each other, one running North-South, the other East-West, and intersecting at their midpoints to form a cross. The strips were used as 1 m wide belt transects for understory sampling, resulting in a typical sampling area of 119 m². Some plots differed from this general layout if, e.g., one strip went over a cliff, into a trail, or into a body of water, but the area sampled for each plot was recorded and standardized densities (e.g., stems/m²) were used in analysis to account for any differences in area sampled among plots and/or years. Canopy closure was measured using an ocular tube and, since 2014, a convex densitometer at the center and at each 10 m interval along every strip.

Weckel et al. (2006) compared a subset of the first sample of these plots in 2004 to an inventory performed in the hemlock forest in 1965 (Bard 1967) and found that the forest understory had substantially changed, with far fewer stems of all woody species below the browse line. Bard's sample was limited to a single strip transect along the MRG Preserve's existing hiking trail within the boundaries of the land owned by MRG in 1967, and Weckel et al. (2006) compared this 1965 survey to data from 5 plots (out of 22) that overlapped with the area sampled by Bard (1967). MRG originally planned to sample these 22 plots every 3 years for understory data. After staff noted that sampling the plots that frequently appeared to be impacting the plots, the schedule was altered to a 5-year schedule after 2010. The final dataset consisted of understory data from 2004, 2007, 2010, 2014, and 2019.

Following Bard's (1967) size categories, we identified every woody stem to species or genus and grouped them into 3 size categories in the north-south and east-west strips: seedlings (arborescent species < 0.3 m tall and having ≥ 1 set of post-cotyledon leaves), saplings (arborescent species 0.3–0.9 m tall), and transgressives (arborescent species > 0.9 m and < 2 m tall). These categories are smaller than those used in most other forest regeneration studies (e.g., the US Forest Service Inventory and Analysis Program), but they were used to maintain comparability with Bard's (1967) study and to ensure early detection of any improvement in woody recovery or species richness. We counted stems of vine species such as grape (*Vitis vulpina*), poison ivy (*Toxicodendron radicans*), oriental bittersweet (*Celastrus orbiculatus*), and Japanese honeysuckle (*Lonicera japonica*), and softer woody stemmed plants such as raspberry (*Rubus occidentalis*), blackberry (*Rubus allegheniensis*), and exotic wineberry (*Rubus phoenicolasius*) that can become substantial components of the understory. Because these species can grow densely, often remain low to the ground, and, in the case of *Rubus* spp., can re-

root, making height measurements difficult and somewhat misleading, they were binned into a “Vine” category with no size definition.

2.4.1. Woody understory analysis

To examine the association and interplay of woody regeneration with deer density, time, canopy cover, and forest type, we built sets of generalized mixed-effect models that examined species richness, stem density of all observed species, and native stem density. Stems of all size classes were pooled for these models; relationships and effects regarding size class and taxa were examined later (see below). We tested four covariates: year zeroed at 2004 (Time); estimated deer/km² for the respective sample year (DD); forest type (Type), which consisted of two categories, coniferous forest (C) or hardwood forest (H); and mean percent closure of all canopy measurement in a plot. There was no deer estimate associated with the 2007 vegetation survey as camera surveys began in 2009. Canopy cover, originally a percentage with little variability, was scaled to a mean of zero and SD of 1 (CC). Pearson correlation coefficients were calculated among Time, DD, and CC; not surprisingly, Time and DD were highly negatively correlated ($r = -0.92$) and thus we did not run parameterizations that included Time and DD in the same model. For each dependent variable (richness, total stem density, and native stem density), we ran the following 12 models using the GLMER function of the lme4 package (Bates et al. 2015) of Program R (R Core Team 2019): an intercept-only model with no fixed-effect covariates; four single-covariate models; 5 two-covariate models of each pair of the four fixed covariates excluding the Time and DD combination; and 2 three-covariate models of each combination of covariates except those including Time and DD together. Richness was modeled as a Poisson-distributed variable using a log link and stem density was modeled as a gamma-distributed variable using an inverse link. All models were run with plot (Plot) as a random intercept.

Overall model fit was assessed by examining quantile–quantile plots of sample residuals versus expected values. Models were evaluated using AIC_c. Models were determined to be “supported” by examining ΔAIC_c values, and w_i 's of the models in each set for obvious drop-offs in support. We averaged the parameter estimates from supported models, weighted by w_i , and explored covariate-parameter relationships by plotting prediction slopes and confidence bands of the averaged parameters. We then examined model-supported associations between species richness and stem density with the supported predictors according to the three stem size classes, as well as species or genera of interest. Any improvement in regeneration would likely be first and

most easily detected in the small seedling class, though after 15 years of deer management we expected to see improvement in the medium sapling class and possibly some improvement – perhaps in only a few particular species – in the larger transgressive class.

The package MuMIn (Barton 2019) was used for model averaging, and the following other packages were used for creating graphs and summarizing results: MASS (Venebles and Ripley 2002), gplots (Warnes et al. 2020), tidyverse (Wickham et al. 2019), and sjplot (Lüdtke 2021).

3. Results

3.1. Harvest and efficiency of DMP

Total deer harvest (all demographics) ranged from 6 deer in 2012 to 35 deer in 2004. There appeared to be a clear shift in harvest after approximately 2010, with a mean total harvest of 29 ± 4.4 (mean $\pm$ SD) deer 2004 – 2010 versus 15 ± 7.0 deer after 2010. Harvest efficiency during the normal season declined (i.e., mean hours/harvest increased) over time, though with substantial variability year-to-year (Fig. 2). The institution of the coordinated post-season harvest in the 2016/2017 provided a 25 – 50% increase in the total annual harvest at approximately half the hours per harvest of the respective regular season. However, this additional harvest still required significant person-hours on the part of MRG staff to obtain (Table 1).

3.2. Deer density

The number of individually identifiable bucks photographed annually from 2009 to 2019 ranged from 12 (in 2015) to 27 (in 2009). SECR estimates of buck densities ranged from 1.3 to 5.0 bucks/km² (Table 4). Generally, the AIC_c-selected best model was an intercept-only model with an exponential detection function, and if model averaging was needed the second-best model was intercept-only with a half-normal function (Table 4). Deer density estimated from camera trap surveys (2009 – 2019) averaged 9.86 ± 2.65 deer/km² (mean $\pm$ SD), ranging from 12.92 deer/km² in 2018 to 5.6 deer/km² in 2019 (Table 5). From 2009 to 2019, density did not trend up or down substantially, though there seemed to be some evidence of 2–3 year cycles of both deer density and harvest (Fig. 2). Thus, the reductions in deer density the DMP achieved occurred in the early years of the program, with efforts since 2011 maintaining this reduction. Back-calculating a herd size from densities to the estimated ‘region’ in SECR (i.e., ‘mask area’) was 21.36

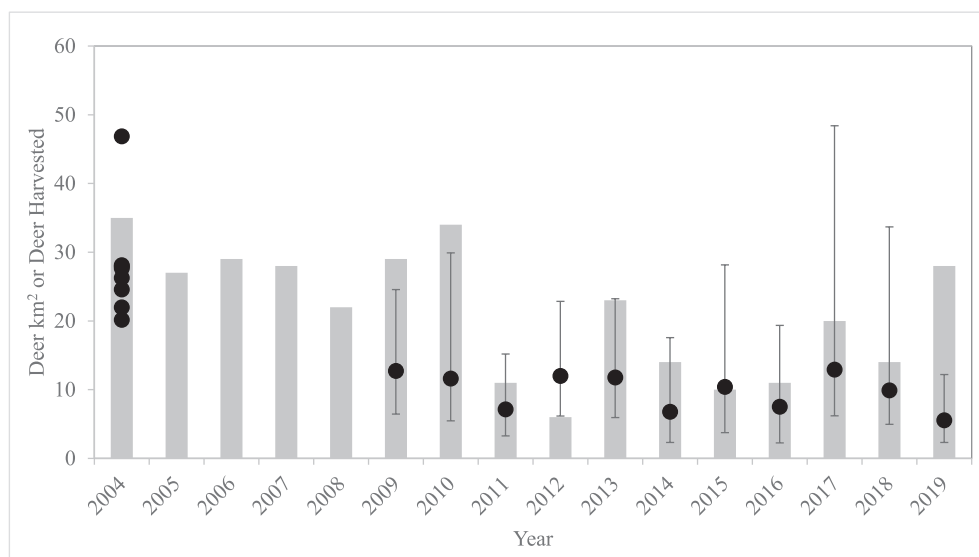


Fig. 2. Total harvest (gray bars) and estimated deer densities (black dots; $\pm$ 95% bootstrapped intervals) during the MRG DMP, Bedford, NY, 2004 – 2019. Deer density in 2004 estimated from regional studies, see Table 2.

Table 4

Supported SECR models, detection functions (E: exponential; HN: half-normal) parameter estimates, and buck density estimates at the Mianus River Gorge Preserve, 2009–2019.

Year	Supported Model(s)	Det. Func.	w_i	D (SE)	$g(0)$ (SE)	σ (SE)	Bucks/km ² (SE)
2009	D(.,g(., σ .)	E	0.85	-3.5293 (0.2439)	-2.3189 (0.3036)	5.5447 (0.1435)	2.93 (0.73)
2010	D(.,g(., σ .)	E	0.96	-3.4672 (0.2796)	0.1267 (0.5632)	4.9061 (0.1421)	3.12 (0.89)
2011	D(.,g(., σ .)	E	0.98	-3.9973 (0.3069)	-2.1499 (0.3633)	5.5694 (0.1667)	1.84 (0.58)
2012	D(.,g(., σ .)	E	0.96	-3.4678 (0.2418)	-1.9766 (0.2871)	5.3313 (0.1338)	3.12 (0.77)
2013	D(.,g(., σ .)	E	0.90	-3.4684 (0.2167)	-1.3534 (0.2657)	5.2272 (0.0961)	3.12 (0.68)
2014	D(.,g(., σ .)	E	0.50	-4.2721 (0.4312)	-3.3533 (0.5088)	5.8250 (0.2814)	1.30 (0.60)
	D(.,g(., σ .)	HN	0.50	-4.4140 (0.4340)	-4.3452 (0.4354)	6.4026 (0.2437)	
2015	D(.,g(., σ .)	E	0.54	-3.8612 (0.4436)	-1.8413 (0.5973)	5.3259 (0.2835)	2.11 (0.97)
	D(.,g(., σ .)	HN	0.46	-3.8578 (0.4297)	-2.9829 (0.4243)	5.8357 (0.2520)	
2016	D(.,g(., σ .)	E	0.91	-4.0276 (0.3454)	-2.3415 (0.3921)	5.6895 (0.2214)	1.78 (0.63)
2017	D(.,g(., σ .)	E	0.96	-2.9921 (0.2883)	-0.5574 (0.4482)	4.5308 (0.1510)	5.02 (1.48)
2018	D(.,g(., σ .)	E	0.82	-3.3168 (0.2544)	0.1367 (0.4033)	4.7039 (0.1084)	3.63 (0.94)
2019	D(.,g(., σ .)	E	0.89	-4.3268 (0.3356)	-1.9211 (0.4012)	5.4680 (0.1824)	1.32 (0.46)

Table 5

Estimated buck, doe, spikes plus 4-points, total deer densities, region size, and estimated regional herd size at the Mianus River Gorge Preserve, Bedford, NY, 2009–2019. Region and herd size based on total mask area generated by spatial capture recapture analysis.

Year	Density (animals/km ²)					SECR Region (km ²)	Regional Herd Size
	Bucks (95% CI)	Does (95% BSI*)	Fawns (95% BSI)	S + 4p (95% BSI)	All deer (95% BSI)		
2009	2.93 (1.77–4.63)	3.96 (1.91–8.13)	2.43 (1.21–4.84)	3.43 (1.54–6.98)	12.75 (6.43–24.58)	29.25	372.88
2010	3.12 (1.74–5.25)	3.26 (1.36–9.35)	2.74 (1.29–6.77)	2.50 (1.06–8.53)	11.62 (5.45–29.90)	13.38	155.43
2011	1.84 (0.96–3.24)	2.15 (0.94–4.83)	1.54 (0.69–3.44)	1.63 (0.66–3.68)	7.16 (3.25–15.19)	35.41	253.50
2012	3.12 (1.89–4.90)	3.64 (1.52–7.90)	2.55 (1.27–5.10)	2.69 (1.48–4.95)	12.00 (6.16–22.85)	20.85	250.18
2013	3.12 (1.99–4.67)	3.37 (1.34–7.10)	2.28 (1.15–5.11)	3.02 (1.46–6.35)	11.79 (5.95–23.22)	17.23	203.15
2014	1.30 (0.55–3.07)	2.25 (0.70–5.99)	1.31 (0.51–3.43)	1.92 (0.54–5.09)	6.78 (2.30–17.58)	33.36	226.19
2015	2.11 (0.89–4.97)	3.95 (1.34–10.90)	2.22 (0.78–5.63)	2.15 (0.73–6.65)	10.42 (2.30–17.58)	16.90	176.10
2016	1.78 (0.63–3.51)	2.52 (0.73–6.09)	1.68 (0.47–4.27)	1.53 (0.42–5.48)	7.51 (2.26–19.35)	25.15	188.91
2017	5.02 (2.79–8.70)	2.63 (1.01–16.35)	2.45 (1.23–8.28)	2.82 (1.16–15.09)	12.92 (6.20–48.42)	8.02	103.66
2018	3.63 (2.14–5.83)	2.65 (1.19–14.07)	2.28 (1.06–5.79)	1.36 (0.55–8.00)	9.92 (4.95–33.70)	11.01	109.21
2019	1.32 (0.65–2.44)	1.72 (0.66–3.98)	1.16 (0.48–2.58)	1.36 (0.52–3.20)	5.56 (2.31–12.19)	24.42	135.77

* BSI: 95% bootstrapped interval.

± 9.02 km² (mean $\pm$ SD), yielding total herd sizes ranging from 103.6 deer in 2017 to 372.3 deer in 2019 living in and around the MRG Preserve (Table 5).

3.3. Woody understory recovery

3.3.1. Modeling results

Several models in both the total stem density set and native-only stem density set were numerically singular, i.e., estimates of the model parameters were on the boundary of the parameter space. When we examined these models we found that it was the random “Plot” effects that were singular with an estimate of 0, suggesting Plot had no discernable effect overall. We removed the random effect and re-ran these models with just their respective fixed effects using a generalized linear model (function GLM) in Program R. Coefficients and standard errors of the other parameters and fixed effects in these models were identical to the original mixed model estimates in all cases and the AIC_c scores were comparable – the GLM models were lower by 4 AIC_c points compared to the GLMER models, i.e., the equivalent of 2 fewer parameters, as expected when removing a single random effect and its residual. In any case, the singular models had poor AIC_c support and

were not considered beyond model selection.

AIC_c-selected best model set for species richness included 2 models (Table 6), the 3-covariate Time + Type + CC model and the 2-covariate Time + CC. These two models had a cumulative w_i of > 0.98 and thus other models had very little relative support. Model-averaged estimates from these two models showed that richness increased with Time ($\beta_{\text{Time}} = 0.0673 \pm 0.0063$ (mean $\pm$ SE)) and was on average 2.5 species greater in the hardwood forest ($\beta_{\text{Type=H}} = 0.2059 \pm 0.0865$), likely due to the greater number of exotic species in the second-growth hardwood forest compared to the old growth hemlock stands (Fig. 3a). CC was present in the two supported models ($\beta_{\text{CC}} = -0.0395 \pm 0.0365$), showing a loss of species as canopy closure increases; however, the range of observed CC was very small in both forest types (mostly between 85% and 95% closure) while the standard error of the coefficient was quite high (Fig. 3b). Deer density did not rank high in species richness models.

There was a general trend of increasing species richness in most plots across the study period, with a mean of 6.0 ± 2.6 (mean $\pm$ SD) species observed per plot in 2004 up to a mean of 17.0 ± 3.3 species per plot in 2019. Native species richness in 2019 was 14.0 ± 2.4 , still over double of 2004. The bulk of the species gain occurred in the seedling class, with more recent improvement in the sapling and vine classes after 2014

Table 6

AIC_c-supported models and parameter estimates for woody plant species richness, stem density of all species, and stem density for native species at the Mianus River Gorge Preserve, Bedford, NY, 2004 – 2019 (Time: years since 2004; Type: forest type, either hardwood (H) or coniferous (C); CC: canopy closure; DD: deer density).

Species Richness	k	ΔAIC_c	w_i	Coefficients (SE)			
				Int	Time	Type = H	CC
Time + Type + CC	5	0.00	0.78	1.7548 (0.0887)	0.0685 (0.0064)	0.2059 (0.0865)	−0.0431 (0.0358)
Time + CC	4	2.68	0.20	1.8555 (0.0799)	0.0677 (0.0063)	x	−0.0257 (0.0356)
Model Average				1.7756 (0.0961)	0.0683 (0.0064)	0.2059 (0.0865)	−0.0395 (0.0365)
Stem Density (All Spp.)	k	ΔAIC_c	w_i	Coefficients (SE)			
DD	4	0	0.50	Int	DD	Type = H	
Type + DD	5	0.68	0.36	−0.3241 (0.1009)	0.1004 (0.0137)	x	
				−0.2763 (0.1113)	0.0999 (0.0136)	−0.0859 (0.0707)	
Model Average				−0.2673 (0.1061)	0.0943 (0.0134)	−0.08606 (0.07084)	
Stem Density (Native Spp.)	k	ΔAIC_c	w_i	Coefficients (SE)			
DD	4	0	0.72	Int	DD	Type = H	
Type + DD	5	2.14	0.25	−0.6673 (0.1415)	0.1678 (0.0211)	x	
				−0.65471 (0.14818)	0.1677 (0.0210)	−0.0242 (0.0737)	
Model Average				−0.6641 (0.1433)	0.1678 (0.0211)	−0.0243 (0.0737)	

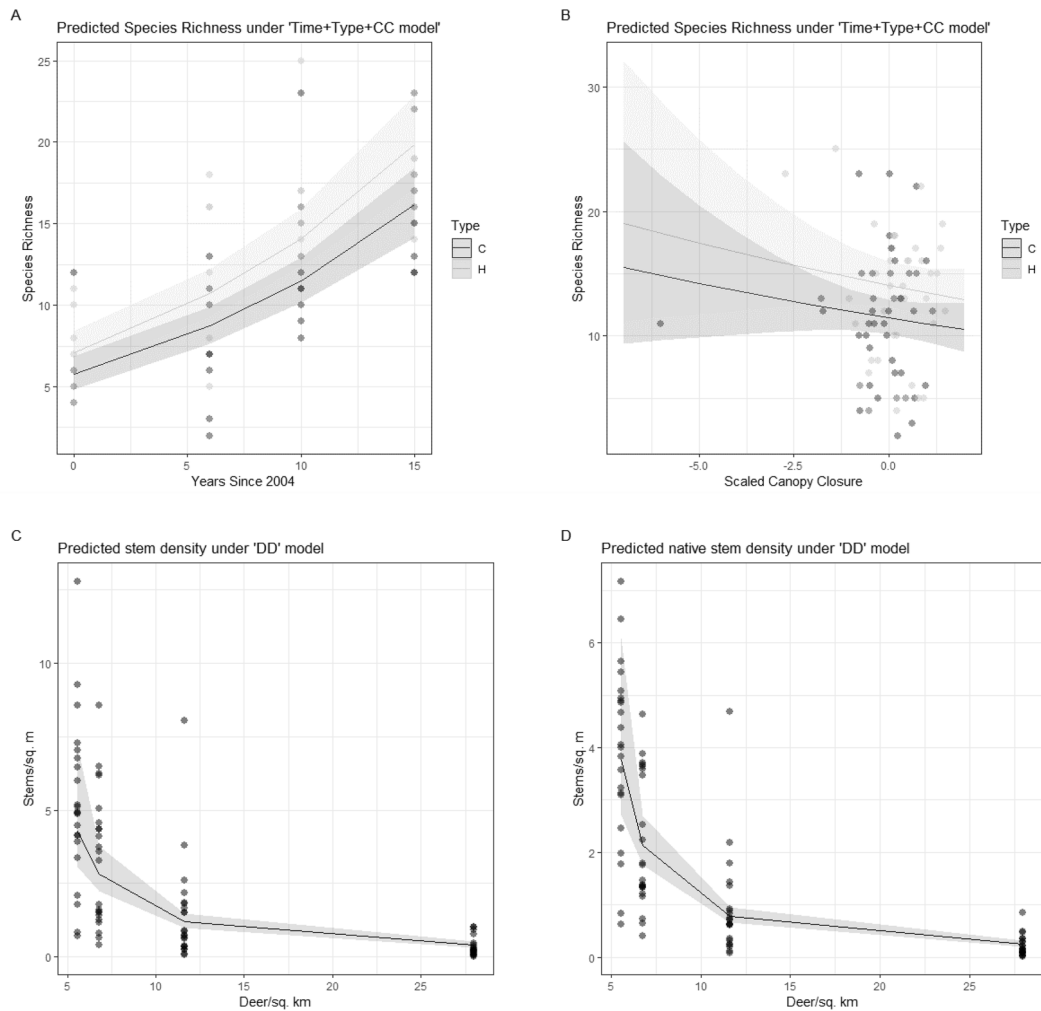


Fig. 3. Prediction curves based on top supported model estimates of (A) species richness versus time; (B) richness versus canopy closure; (C) total stem density versus deer density; and (D) native stem density versus deer density. “C” or “H” in richness figure legends refers to hardwood or coniferous stands. Dots indicate actual measurements at individual plots, shaded bands indicate 95% confidence bands.

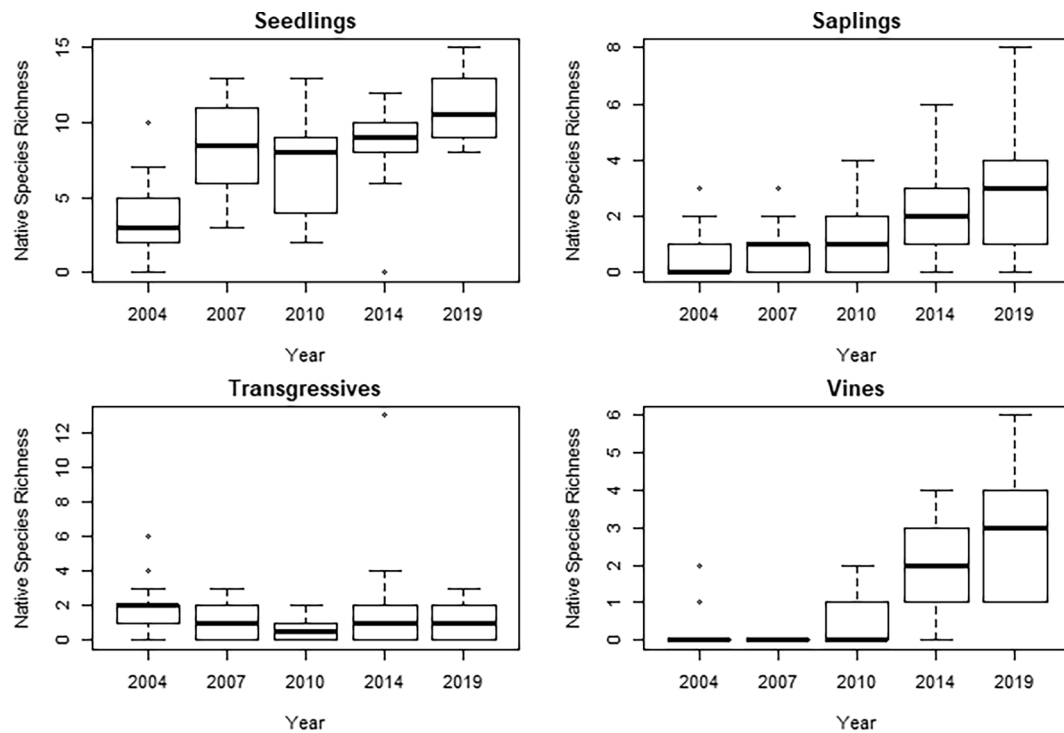


Fig. 4. Native woody species richness across vegetation plots at the Mianus River Gorge Preserve, Bedford, NY, 2004 – 2019. Seedlings: woody stems with 2 post-cotyledon leaves < 30 cm tall; Saplings: woody stems 30 – 90 cm; transgressives: 90–200 cm.

(Fig. 4). There was virtually no change in richness in the transgressive class.

The model parameterizations that were supported for total stem density and native stem density were identical; as with richness the two top models in these sets contained the majority of support (All stems: $w_{1+2} > 0.86$; native stems: $w_{1+2} > 0.97$) with noticeable “drop-offs” in support at the third-ranked model and below (Table 6). The top model in both sets was the single-covariate DD model, while the second was DD + Type. In fact, the 5 models that included DD were the top 5 models. Because the top model was the single-predictor model it is likely that the second supported model (DD + Type) was ranked high simply because it included DD. Regardless, model averaged estimates of β_{DD} did not differ meaningfully from the estimate of β_{DD} from the top model alone so we proceeded with using model-averaged predictions (Table 6).

Stem densities were higher when deer density was low (All stem density: $\beta_{DD} = 0.0960 \pm 0.0137$; Native stem density: $\beta_{DD} = 0.1706 \pm 0.0215$). While all plots had few stems at high deer densities, at low deer density some plots remained at low stem density while others appeared to respond strongly (Fig. 3c and 3d). Forest type had limited real effect except perhaps at the lowest deer densities where hardwood forests may have slightly more stems per meter; as mentioned above, Type appeared in the second-ranked model primarily because it was paired with DD.

3.3.2. Species composition changes

Weckel et al. (2006) compared species-specific stem densities between the 1965 (Bard 1967) and their own 2004 samples from the 5 plots that overlapped the 1965 strip transect. Adding the 2019 data from these 5 plots to this series, we observed a few patterns. First, black birch (*Betula lenta*), red (*Acer rubrum*) and sugar maple (*Acer saccharum*), and American beech (*Fagus grandifolia*) increased markedly in 2019 compared to both prior surveys. For black birch this occurred across all three size classes. Beech stems improved in the sapling and transgressive class but beech seedlings were not observed in substantially greater densities in 2019 compared to prior years. Red maple did not improve in the transgressive class but was more common in the seedling and sapling classes in 2019 than earlier. Sugar maple was similar to red but

transgressives improved somewhat after 2004 (Fig. 5).

Because the 1965 to 2004 comparison was limited to only 5 plots, we also compared the species assemblage across 2004 to 2019 using all 22 plots and all 5 surveys (2004, 2007, 2010, 2014, and 2019). Many native species increased dramatically in the smaller size classes but almost no important native species exhibited improvements in the transgressive class (Fig. 6). The exceptions were black birch, American beech, and maple-leaf viburnum that increased an average of 1.4, 1.2, and 0.06 transgressive stems/m annually over the study period, respectively. Eastern hemlock, a priority management species for MRG, was rare, showing some improvement in the seedling class but little beyond that.

4. Discussion

4.1. Woody vegetation recovery

The MRG Preserve’s woody understory community, in terms of overall species richness and stem density, has improved from the start of the DMP in 2004 through the present. However, the majority of the regeneration in native species occurred in the shortest (<30 cm) stem class only. Thus the ultimate assessment of the DMP in terms of facilitating understory regeneration remains somewhat vague. Either deer management as implemented by MRG has a measurable but ultimately inadequate effect on tree and shrub regeneration, allowing very young stems to survive a few years longer but still never grow above 1 m, or the DMP has allowed the understory to begin to recover and continued management will gradually lead to improving transition rates from the short size classes into the taller. Monitoring will continue if deer management is to continue, and presumably in the future a similar assessment can provide more clarity.

Another complication with assessing the DMP’s impact on understory regeneration arises from the fact that time and deer density were correlated, i.e., deer density declined as deer management proceeded. This is unsurprising, and in fact was the goal, but it made determining whether the observed improvements in vegetation are due to DMP activities versus recovery over time for other reasons, e.g., natural forest

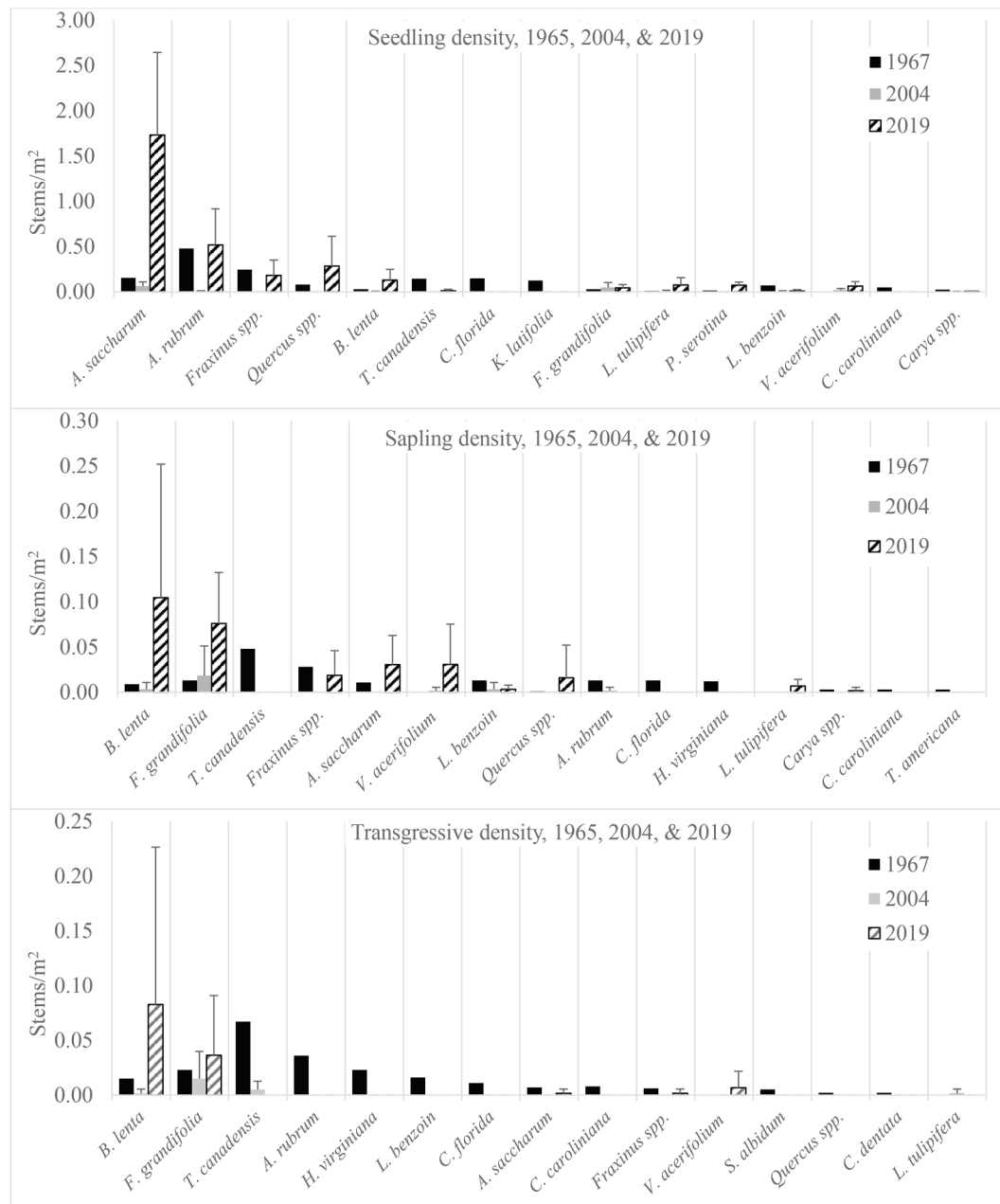


Fig. 5. Stem density (mean stems/m² + SD) of the top 15 densest woody species measured at the Mianus River Gorge Preserve in 1965, 2004, and 2019 for three size classes: seedlings (<30 cm tall), saplings (30–90 cm tall), and transgressives (90–200 cm tall). Only plots that overlapped with the 1965 transect survey are shown here (n = 5).

maturation and succession; a long-term, gradual recovery from the region's agricultural history; or other management activities implemented on the Preserve. Modeling Time and DD in separate models and comparing them via AIC_c addressed this issue to some extent, and in the case of species richness it appears that Time was more meaningful whereas for stem density DD was a better predictor. Without direct control-versus-treatment comparisons (i.e., plots located in similar locations without deer management over the same period) we must look for the most proximate and likely reason for the measured recovery in the understory. Many other studies have documented the long-term impacts of super-abundant white-tailed deer in a variety of forest types (e.g., boreal forest: Tremblay et al. 2007; eastern deciduous forests: Bradshaw and Waller 2016, Royo and Stout 2019; hemlock forests: Salk et al. 2011, Alverson et al. 2019, Parikh and Webster 2019; sub-urbia: Duguay and Farfaras 2011, Blossey et al. 2019; and northern

forests regionally: Miller and McGill, 2019, Russel et al. 2017). Overall, it seems unlikely that the 40-year trend of biodiversity and understory degradation observed at the MRG Preserve prior to deer management (Weckel et al. 2006) would have reversed course simultaneously but unrelatedly with the implementation of an intensive DMP.

In addition, many of the factors that affect forest understory composition and dynamics differ spatially in MRG Preserve, namely, by forest type. The hardwood areas of the Preserve are all second-growth, post-agricultural forests with relatively flat topology, less woody debris (Russel et al. 2017), homogenized soil with potentially degraded mycorrhizal communities (O'Brien et al. 2011, Karpati et al. 2011), and more exotic plants and earthworms (Fischelli et al. 2012, Walling 2014, (Dávalos et al., 2015)) – these factors all play a role in woody regeneration and overall forest health. Yet these differences between the more pristine old-growth coniferous plots and the second-growth hardwood

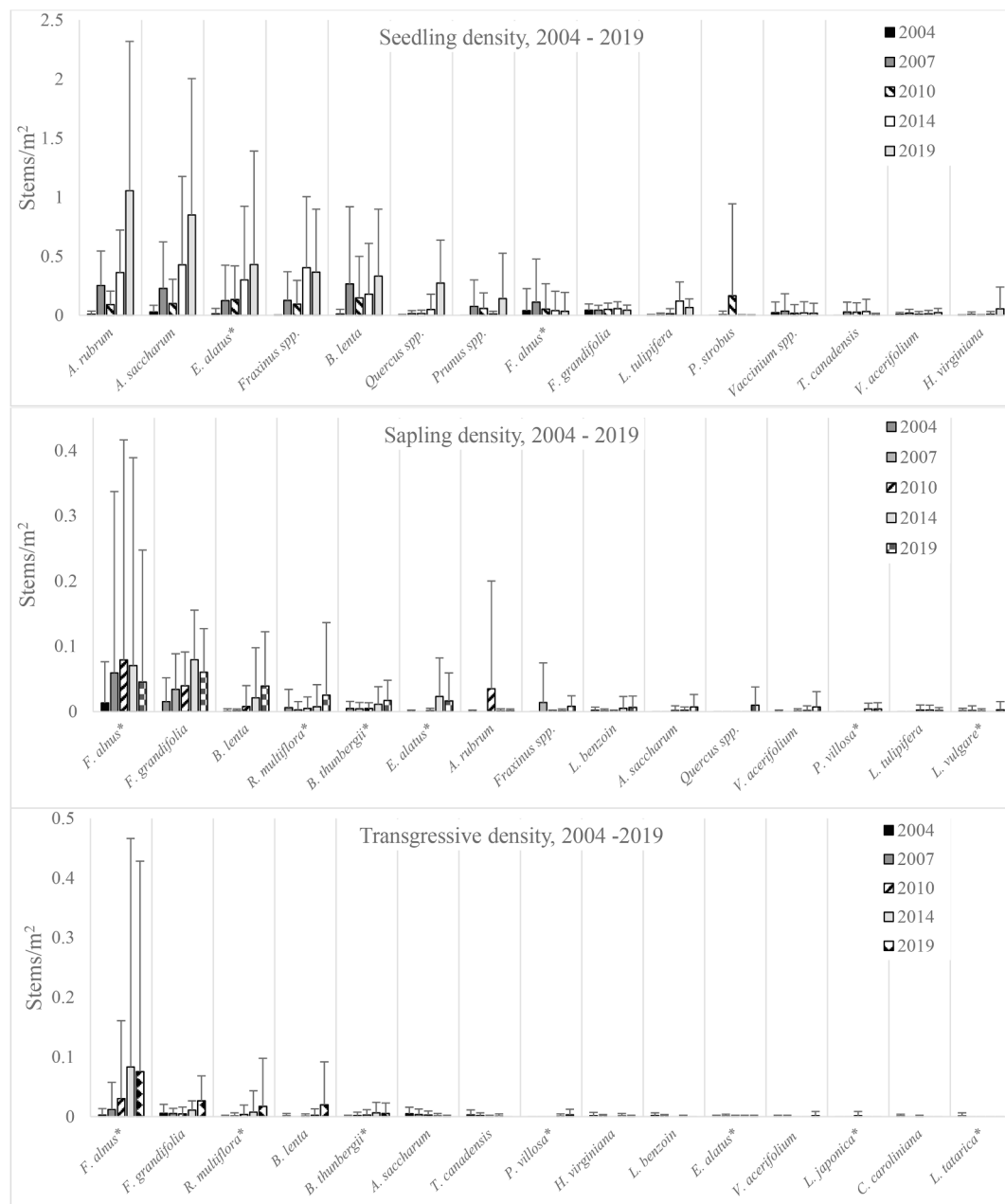


Fig. 6. Stem density (stems/m² + SD) of the top 15 densest woody species measured at the Mianus River Gorge Preserve in 2004, 2007, 2010, 2014, and 2019 for three size classes: seedlings (<30 cm tall), saplings (30–90 cm tall), and transgressives (90–200 cm tall). Asterisks after species names indicate invasive (exotic) species.

plots did not emerge as important via the Type factor during modeling. Future research should examine these alternative factors more directly; even if deer are the main driver of understory regeneration, other variables still play important roles and may themselves need to be managed to improve the situation.

Most other vegetative studies ignore very small seedlings and count only those woody stems above a minimum height for ease of sampling. Typically, protocols follow the US Forest Service Forest Inventory and Analysis (USFIA) stem categories, where hardwood seedlings are defined as small trees > 30.48 cm but no thicker than 2.54 cm dbh. Stems from both our sapling and transgressive categories would be classified as seedlings in the USFIA protocol, so we pooled these classes and compare them with USFIA-seedling densities from other studies to determine if the observed improvements are ecologically significant. Miller and McGill (2019) used a USFIA-seedling density of 0.25 stems/

m² as a minimum for future overstory recruitment. Mean stem densities at the MRG in 2019, pooled across saplings and transgressives and excluding vines, reached 0.41 stems/m², but when exotic species (e.g., glossy buckthorn, winged euonymus, oriental photinia) were removed the mean fell to 0.20 stems/m², failing to reach the minimum guideline. This total still includes native shrubs such as spicebush (*Lindera benzoin*) and witch-hazel (*Hamamelis virginiana*) – removing these to examine just those species that can reach canopy height lowers the mean further.

While native stem density did not meet this minimum threshold, it has increased since 2004 (i.e., increased as deer became less abundant), from 0.03 to 0.20 USFIA-seedlings/m². This measurable, if not-yet adequate, trend suggests that continued deer management should yield positive results, as long as the DMP can maintain effective harvest rates. While one might expect a 15-year time frame to be an adequate period within which to obtain a definitive conclusion on the efficacy of a

DMP, in some cases the legacy effects of high deer densities can extend 20 years or more after deer numbers have been reduced (Nuttall et al. 2011, Nuttall et al. 2014). Further, Weckel and Rockwell (2013) modeled the deer population in the MRG Preserve and concluded that pushing deer densities down to a target of 5–7 deer/km² (based on Tilghman's (1989) recommendation) is only possible in a limited set of circumstances, and even then, could take 20–30 years. Other studies examining understory regeneration during deer management found similar short-term-only recoveries after four (Simard et al., 2013), eight (Schmitt et al. 2020), and 13 (Tanentzap et al. 2011) years. If reaching target deer densities takes over a decade, and legacy effects can remain for even longer, then the observed improvements in species richness and stem density over the first 15 years may be a hopeful sign, and the DMP and the associated monitoring efforts should continue. This multi-decade minimum is admittedly a frustrating prospect for managers looking for clear guidance on where to spend their limited resources.

Species that did recover shared some characteristics. First, the native species that showed the greatest recovery in the smaller classes were those that are well represented in the overstory – red maple, black birch, and sugar maple (Mianus River Gorge, unpub. data) — and thus could produce the most propagules year-after-year. The stark exception here was eastern hemlock, which was by far the most common overstory tree in 2004 but failed to regenerate meaningfully in any size class. In the transgressive class, four of the six most improved species were non-native invasive species: glossy buckthorn, multiflora rose, Japanese barberry, and oriental bittersweet. The two natives in this group were American beech and black birch, the former able to reproduce via clonal root-sprouts.

4.2. Harvest efficiency and deer density

If the 2004 region-based estimate of 28.0 deer/km² was accurate, then the DMP caused a substantial reduction in deer density. It is unfortunate that no MRG-specific unharvested deer estimate was available; however, data from nearby and similar preserves, parks, and other protected open spaces all report deer densities above 20 deer/km² (Table 2) and there is little reason to suspect the MRG had uniquely low deer density compared to similar forests nearby. From a 2004 density of 28.0 deer/km² to a 9.34 deer/km² (the mean of our estimates since 2011 when harvest seemed to drop) is a 67% reduction. Clearly there is uncertainty around these estimates, but using the minimum estimated density from our local sources (Table 2) of 20.2 (Brash et al. 2004) and the maximum density we estimated of 12.92 (in 2017) would still indicate a 37% reduction. Thus, we are confident that deer management has substantially lowered deer density in the MRG. However, across the 2009 – 2019 period, when the camera-based deer estimates were calculated, density declined only slightly (Fig. 2). It appears the reduction from 28 deer/km² to ~10 deer/km² occurred primarily in the first 7 years of the program when high harvest numbers (>20 deer/season) were maintained.

After modeling deer population dynamics and hunter efficiency of the MRG DMP after 6 years, Weckel and Rockwell (2013) concluded that MRG, and archery-only DMP's in general, would be hard-pressed to attain meaningful deer reductions except under certain circumstances; namely, low rates of immigration and moderate to high hunter efficiencies. In suburban areas with non-firearm DMPs, meaningful deer reductions (i.e., reaching densities that are thought to result in understory regeneration) were largely unachievable in many instances (Bou-langer and Curtis 2016, Blossey et al. 2019, Williams et al. 2013). That the MRG DMP was able to achieve a large reduction, similar or better than rifle-based DMPs (Williams et al. 2013, Weckel and Rockwell, 2013) is notable and, to some extent, due to the complete lack of hunting at the MRG prior to the implementation of the DMP: deer in the MRG Preserve were largely naïve to hunting, so recreational hunting remained effective enough to drastically reduce the deer population for several years. However, as deer became less abundant and those that

remained were “educated” (Williams et al. 2013), it was necessary to establish an additional post-season harvest using crossbows, bait, and lights under a DDP as recreational hunting under normal season regulations became less efficient. Many other DMPs have encountered the same declining efficiency of recreational hunting (Kilgo et al. 1998, Williams et al. 2008, Kilpatrick et al. 2004). Blossey et al. (2019) ultimately removed the recreational component of their DMP activities and shifted entirely to a coordinated DDP program. This conversion may be necessary for the MRG to enact if deer densities are to be lowered below their current estimated levels of around 10 deer/km².

The reliance on recreational hunters also brings up another challenge in that the goals of managers and hunters will likely diverge. When deer were at very high density (e.g., >20 deer/km²), hunters and land managers concerned with forest health generally agree that numbers should be reduced. From many hunters' perspectives, these high densities were leading to lower “quality” trophy bucks (Weckel et al. 2011b, Harper et al. 2012). However, if deer are reduced to the very low targets sought by land managers (e.g., 2–10 deer/km² depending on management goals), it seems inevitable that hunting will become very challenging, which in turn could lead to lowered enthusiasm among hunters that DMPs rely on (Miller and Graefe 2001, Weckel et al. 2011b). In other words, the density of deer that provides, or is perceived by hunters to provide, optimal trophy hunting in the northeast is probably a good deal higher than what is necessary for long-term forest health and biodiversity conservation (deCalesta and Stout 1997, Weckel et al. 2011b, Weckel and Rockwell 2013, Blossey et al. 2019). Currently, there is a moderate correlation between deer density and average hours/harvested deer in the MRG DMP ($r = -0.40$). Lower deer numbers may be making harvest more difficult but other factors – such as annual mast crop and/or annual variations in weather and precipitation – likely affect annual hunting success too.

The impact of deer varied by tree species due to browsing preference of deer and the growth potential and other ecological factors of the different tree species. Other research has acknowledged the overall complexity and the site-specificity of deer-habitat relationships, but as a guideline many have recommended deer density around 2–7 deer/km² as a management target to facilitate the regeneration of all species in forested (i.e., non-agricultural areas with generally closed canopy) lands (Tilghman 1989, deCalesta and Stout 1997, Horsley et al. 2003). A higher target deer density of around 8–11 deer/km² could be adequate to allow regeneration of a smaller but still meaningful set of plants (Williams et al. 2013, deCalesta and Eckley 2019). The MRG DMP seems to have reached or has approached this latter target; however, MRG is particularly concerned with the viability and persistence of its old-growth hemlock stands. Deer herbivory is an important factor, but one of many influencing hemlock persistence in the region (Mladenoff and Stearns 1993), most notably the destructive hemlock woolly adelgid (*Adelges tsugae*), for which the MRG has implemented a comprehensive control program. Precise estimates of what deer density can allow for the regeneration of hemlock are scant, but those that do exist are low: ~1–6 deer/km² (Alverson et al. 1988, Rooney et al. 2000). Reaching and maintaining this target seems unlikely without a dramatic improvement in harvest efficiency or some other major ecological shift, e.g., the return of eastern wolves (*Canis lupus lycaon*) or cougars (*Felis concolor*) to the region (Enck and Brown 2002, Nilsen et al. 2007, Blossey et al. 2019).

5. Conclusions and recommendations for the MRG DMP

Our 15-year assessment of the MRG DMP suggests that deer management is having a gradual positive effect and thus, in general, the program should continue. Overall the declining trend from 1965 to 2004 (Weckel et al. 2006) has been reversed, with greater species diversity and abundance in small/young woody stems. In the understory of a mature forest, seedlings can grow extremely slowly, taking 20–125 years or more to grow above the browse line for red oaks (Blossey 2019) and 23–150 years or more for eastern hemlock (Rooney and Waller 2003).

Thus the expectation of dramatic improvement after “only” 15 years may be overly optimistic, and the more modest improvements we measured may be as good a sign as one could expect. However, how can managers expect to stave off herbivory if recruitment of small trees takes two decades or longer? Deer densities must reach extremely low levels for this to occur and must be maintained for a long time, and this extreme reduction is probably unrealistic via recreational hunting alone.

Thus, more work needs to be done on strategies to improve harvest efficiency. Even in the post-season harvest, where bait, crossbows, and night hunting is permitted under the DDP, harvest rate still sits around 30 h/harvest. Weckel and Rockwell (2013) recommended 20–25 female deer be harvested annually at the MRG Preserve to continue to drive the population density down. This would on average represent approximately 750 h of hunting annually even at the better post-season harvest rate. Firearms would normally be an obvious next step; however, extensive public outreach was required to “win over” the community to the archery-based plan of the MRG DMP and we suspect firearm use would be met with even more hesitancy by the community. There are three towns in the vicinity of the MRG Preserve with firearm ordinances that would need to approve a rifle or shotgun cull, and public outcry has already severely impacted a USDA-implemented cull at a nearby Preserve (D. Begley-Miller, Director of Science and Stewardship, Teatown Lake Reservation).

It is imperative that deer management be paired with other management efforts such as canopy manipulation, deer exclosures, and invasive species management that will facilitate more widespread and rapid regeneration. In more pristine settings, canopy manipulation and other standard forestry practices paired with exclosures could be powerful tools towards this end. However, in a suburban landscape like the one the MRG Preserve lies within, opening the canopy often facilitates the spread and growth of invasive species. Other factors are further complicating and confounding forest managers. Invasive earthworms are degrading soil and litter conditions (Dávalos et al., 2015) while several exotic tree parasites have removed and are removing even mature trees, starting with Chestnut blight (*Cryphonectria parasitica*) and Dutch elm disease (*Ophiostoma* spp.) in the 20th century up to current threats such as emerald ash borer (*Agrilus planipennis*), hemlock woolly adelgid (*Adelges tsugae*), beech bark disease (caused by the insect *Cryptococcus fagisuga* and subsequent infection by the fungi *Neovectria faginata*), oak wilt (*Bretziella fagacearum*), and spotted lanternfly (*Lycorma delicatula*), just to name a few. The lack of fire as a natural disturbance cycle and as a permitted management tool in the municipalities involved has further affected the dynamics of the environment within which many of these species evolved. Lastly, even though coyotes (*Canis latrans*), bobcats (*Lynx rufus*), and most recently, black bear (*Ursus americanus*), have arrived or returned to the Preserve, top predators (e. g., wolves or cougars) that have the potential to consistently take adult deer and thus meaningfully impact deer numbers are vital components of the ecosystem that are currently missing.

The process that the MRG DMP and other DMPs hope to enact – hunting will lower deer numbers, lower deer numbers will in turn lessen browse pressure, and decreased herbivory will lead to vegetative recovery – seems to have occurred in the 15 years of the MRG DMP. However, the effects have been muted, with deer density being reduced but not to target levels, and vegetative recovery occurring in small stems and limited in that only a few native species (black birch, American beech, and, to some extent, maple-leaf viburnum) reaching 30 cm or more in height – far less than was hoped for by the Preserve’s managers. Additionally, the level of effort and resources devoted to the DMP has been substantial and scaling up such a program to hundreds of square miles (e.g., to the County level) across parcels of varied land use, ownership type, and access would require tremendous resources. Overall, one must conclude that improving forest regeneration via deer management is, at best, exceedingly difficult. Circumstances such as management being limited to archery only, working across numerous non-contiguous patches of natural land within a matrix of human

development, complimentary management options such as burning or canopy manipulation being risky or prohibited may then raise the bar from “difficult” to “virtually impossible.” MRG’s results are encouraging, but have required 15 years of intensive management and, at best, indicate that another ≈15 years of commitment is necessary. The lack of hemlock regeneration to the larger size classes is probably the most discouraging result relative to MRG’s mission.

Managers at the MRG should continue to experiment with new strategies to increase harvest rate, while further research on several fronts could provide important information. It would be prudent to validate the modified-Weckel et al. (2011a) deer estimator with a more vetted method, such as a mark-recapture or mark-resight study that directly censuses all demographic groups. Information on deer immigration and movements, particularly of does and in response to hunting pressure, would help managers set harvest targets and make predictions on the potential of the DMP overall. Concurrent research on invasive species ecology and management is also vital. Lastly, vegetative monitoring should continue and a final assessment of the DMP should be made in another 10–15 years.

CRedit authorship contribution statement

Christopher Nagy: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Software, Supervision, Visualization, Writing – original draft. **Chloe Ng:** Formal analysis, Investigation, Software, Visualization, Writing – original draft. **Norman Veverka:** Investigation, Project administration, Resources, Writing – review & editing. **Mark Weckel:** Conceptualization, Data curation, Funding acquisition, Investigation, Methodology, Project administration, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

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